SEAL_original | Detailed computational workflow

Exact historical sampling and learning logic used by the CUDA-enabled implementation

DATA & SAMPLING

1 Load network

MAT file → sparse adjacency net
One empirical food web per run
Seed controls Python, NumPy and PyTorch

2 Split observed links

Use triu(net, k=1)
Shuffle positive links
ceil(0.90 × L) train; remainder test

3 Sample unknown pairs

random.randint over i < j and net[i,j] = 0
Match positive counts → 1:1 classes
Historical sampler is preserved unchanged

4 Build observed graph A

Copy net
Mask every held-out positive link
in both directions; eliminate zeros

for every sampled pair

SUBGRAPH ENCODING

5 Candidate pair (u, v)

Process train+/train− and test+/test−
Graph target label: link = 1, non-link = 0
Subgraphs extracted in parallel on CPU

6 Extract 1-hop subgraph

Collect neighbors around u and v
Move target nodes to positions 0 and 1
Induce a NetworkX graph from A

7 Apply structural encoding

Compute DRNL labels from shortest paths
Remove the target edge inside the subgraph
No Node2Vec; no ecological node attributes

8 Create graph datasets

One-hot DRNL node tags in GNNGraph
Balanced positive/negative examples
10% of training graphs reserved for validation

train on graph examples

LEARNING & EVALUATION

9 DGCNN representation

Graph-convolution widths: 32, 32, 32, 1
Concatenate latent layers
SortPooling: 60th percentile, minimum k = 10

10 Neural classifier

1D convolution → max-pool → 1D convolution
MLP hidden size 128 with dropout
Two-class log-softmax output

11 Optimise & select epoch

Adam, learning rate 10^{-4}; batch size 50
50 epochs; negative log-likelihood loss
Retain predictions at minimum validation loss

12 Score held-out links

P(link) from class-1 probability
Positive prediction when score > 0.5
ROC-AUC, PR-AUC, F1
Precision, Recall, MCC, TSS

REPRODUCIBILITY NOTE The CUDA layer changes execution hardware only; sampling, splits, subgraphs, labels, model and metrics retain SEAL_original behavior.

Current campaign: 290 food webs × 10 runs · experiment seeds 12345–12354 reused across food webs · no Tukey filtering.

Sources: Main.py · util_functions.py · pytorch_DGCNN · run_foodweb_seal.py